

Zadanie 20 [3pkt] - TYP 3.4 (metoda 1 i 2)

Rozwiąż nierówność: $2x^5 - x^4 - 2x^3 + x^2 - 4x + 2 < 0$

$$\underline{2x^5} - x^4 - \underline{2x^3} + x^2 - \underline{4x} + 2 < 0$$

Grupujemy:

$$2x(x^4 - x^2 - 2) - 1(x^4 - x^2 - 2) < 0$$

$$(x^4 - x^2 - 2)(2x - 1) < 0$$

$$x^4 - x^2 - 2 = 0$$

$$2x - 1 = 0$$

Podstawiamy $t = x^2$:

$$\begin{aligned} 2x &= 1 \\ x_1 &= \frac{1}{2} \end{aligned}$$

$$t^2 - t - 2 = 0$$

$$\Delta = b^2 - 4 \cdot a \cdot c$$

$$x_1 = \frac{-b - \sqrt{\Delta}}{2a} \quad x_2 = \frac{-b + \sqrt{\Delta}}{2a}$$

$$\Delta = 1 + 8 = 9$$

$$\sqrt{\Delta} = 3$$

$$x_1 = \frac{1-3}{2} = -1$$

$$x_2 = \frac{1+3}{2} = 2$$

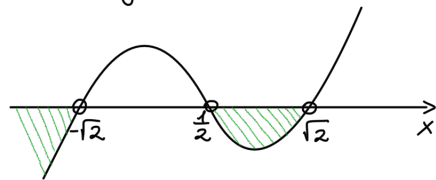
Wracając do $t = x^2$

$$-1 \neq x^2$$

$$2 = x^2$$

$$\boxed{x_3 = \sqrt{2} \quad x_4 = -\sqrt{2}}$$

Zatem wykres funkcji będzie mieć postać:



$$\text{Odp: } x \in (-\infty; -\sqrt{2}) \cup (\frac{1}{2}; \sqrt{2}).$$